Comparative Analogy Reasoning in Transformer-Based Language Models

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Abstract

Analogical reasoning represents a core component of human intelligence — the ability to identify structured relationships between concepts and apply these mappings to novel contexts. As artificial intelligence systems increasingly exhibit language understanding capabilities, assessing whether they can perform this type of relational reasoning has become a critical question in the study of machine intelligence. Recent transformer-based architectures, such as BERT, RoBERTa, DistilBERT, GPT-2, and T5, have achieved success in linguistic and contextual comprehension tasks, yet their capacity for analogical reasoning remains insufficiently explored. This research presents a unified framework for evaluating analogical reasoning across different transformer architectures. A curated set of analogy-based prompts is used to probe each model's ability to infer relational patterns (e.g., "Pilot is to Airline as Sailor is to [MASK]"). The findings aim to contribute to the interpretability of language models, shedding light on how variations in transformer design influence analogical reasoning accuracy.

Keywords: Analogical reasoning, Transformers, NLP, LLM

1 Introduction

1.1 What is analogical reasoning?

An *analogy* [1] is a comparison between two ideas, objects, events, or systems that emphasizes the ways in which they are similar in structure or function. At its core, an analogy highlights a relational correspondence between elements in different domains. A classic way to express an analogy is in the form "A is to B as C is to D," where the relationship between A and B mirrors the relationship between C and D. For example, in the analogy "Bird is to Nest as Bee is to Hive," the relationship of the bird to its nest is similar to the relationship of the bee to its hive, in terms of dwelling and shelter. Analogies are not limited to objects; they can involve abstract concepts, actions, or processes, such as "Electric current is to wire as water flow is to pipe," where the analogy emphasizes the structural similarity between electricity flow and water flow. By identifying these patterns of similarity, analogies allow humans to connect familiar and unfamiliar domains, making sense of new situations through known relationships. *Analogical reasoning* is any type of thinking that can create or complete an analogy through reasoning.

Analogical reasoning [2][3][4] is central to how humans form conceptual understanding, transfer knowledge, and solve novel problems. Analogical reasoning is the cognitive process by which humans create, understand, and manipulate analogies. It is the ability to perceive relational patterns in one context and apply them to another, potentially unfamiliar, context. This type of reasoning is fundamental to learning, problem-solving, and conceptual understanding. For instance, a student who understands how planets orbit the sun in a solar system can analogically reason about the structure of an atom, mapping the relationship between planets and the sun onto electrons and the nucleus. Similarly, in everyday life, people use analogical reasoning when they infer solutions to novel problems by relating them to previously encountered scenarios. For example, someone who knows how to navigate a busy street using traffic rules may apply similar logic when navigating a crowded shopping mall. Analogical reasoning goes beyond simple feature matching; it requires understanding **relationships and patterns** rather than just individual attributes. It involves identifying relevant similarities, ignoring irrelevant details, and mapping relational structures across domains.

In cognitive science [5], analogical reasoning is recognized as a cornerstone of higher-order thinking. It allows humans to generalize knowledge, transfer skills, and generate creative solutions. By recognizing patterns of similarity and relational correspondence, people can make predictions, explain phenomena, and solve problems that they have not encountered before. For example, scientists often use analogical reasoning to formulate hypotheses, such as comparing the flow of water to the flow of electricity, which can lead to deeper insights and novel discoveries. Philosophers, educators, and psychologists have long emphasized the role of analogical reasoning in understanding complex ideas and facilitating learning [6].

1.1.1 Types of Analogies

Analogies can take several different forms, each highlighting a specific kind of relationship between elements. Understanding these types is essential for studying analogical reasoning, as different analogical tasks test different cognitive or semantic capabilities. Three common types of analogies are functional analogies, taxonomic analogies, and role–object analogies [7].

1. **Functional Analogies**
   Functional analogies focus on the relationship of an agent to the object or tool that it operates or controls. In other words, they capture how one entity uses, interacts with, or is dependent upon another in terms of function. For example, consider the analogy "Pilot is to Plane as Sailor is to Ship." Here, the relationship is based on the functional role of the agent in controlling or operating the object. A pilot flies a plane, and similarly, a sailor navigates or operates a ship. Functional analogies emphasize understanding the purpose, use, or function of objects in context and mapping those relationships to another domain. Solving these analogies requires recognizing the role of the first element (Pilot) in relation to the second (Plane) and applying the same functional relationship to identify the missing element (Ship).
2. **Taxonomic Analogies**
   Taxonomic analogies focus on categorical or hierarchical relationships between entities. These analogies examine whether the reasoner can recognize that two items belong to a broader category and then apply that same categorization logic to another set. For example, the analogy "Sparrow is to Bird as Salmon is to Fish" illustrates a taxonomic relationship. Here, sparrows are a type of bird, and salmon are a type of fish. The analogy tests the ability to understand category membership and the hierarchical relationship between specific instances (Sparrow, Salmon) and their general classes (Bird, Fish). Taxonomic analogies are important for assessing whether a model or individual can organize knowledge conceptually and reason about class–subclass relationships, which are fundamental in learning and generalization.
3. **Role–Object Analogies**
   Role–object analogies describe the relationship between a person or agent and the environment, object, or context in which they perform their role. These analogies often examine professional, occupational, or contextual associations. For example, "Chef is to Kitchen as Teacher is to Classroom" demonstrates a role–object analogy. A chef's work is associated with the kitchen, and a teacher's work is associated with the classroom. Solving these analogies requires understanding the typical setting or object connected to a person's role and then mapping that relationship to another domain. Role–object analogies emphasize relational reasoning about social, functional, or environmental context rather than pure category membership or functional interaction with tools.

1.2 Analogical Reasoning in Natural Language Processing

In natural language processing (NLP) [8][9], analogous tasks serve as a diagnostic tool for evaluating semantic understanding—for example, identifying that "Pilot is to Airline as a Sailor is to Ship." This is indicative of reasoning capability given some degree of context, while also testing whether the model is capable of relational understanding. The context refers to the first part of the sentence that does not contain the missing analogy, for example the "Pilot is to Airline" part. The model's capability in being able to accurately determine the analogy is stringent on whether is is able to understand the **relational structure** of the context, recognize that it should use that structure to complete the missing analogy, and accurately follow suit.

The model must recognize the type of relationship—such as occupation to associated workplace—and use that relational understanding to generate the correct answer. This requires reasoning, generalization, and abstraction, all of which are central to analogical thinking.

Evaluating analogical reasoning in NLP serves several purposes. First, it tests whether models can go beyond surface-level associations and truly understand relational semantics. Second, it measures the ability of models to generalize learned knowledge to new situations, reflecting a form of cognitive flexibility. Third, it provides insights into the limitations of current NLP systems, helping researchers design better models, training procedures, and datasets that capture relational reasoning. By studying analogical reasoning, researchers gain a dual perspective: a deeper understanding of human cognition and a framework for assessing artificial intelligence systems in tasks that require complex relational understanding.

Overall, analogical reasoning is both a fundamental cognitive skill and a critical benchmark in evaluating intelligent systems. Analogies themselves are structured comparisons that highlight relational similarities between domains, while analogical reasoning is the process of identifying, mapping, and applying those relationships to novel contexts. This reasoning enables humans—and potentially machines—to transfer knowledge, solve problems creatively, and develop a deeper conceptual understanding of the world.

1.2.1 Transformer language models and analogies

Transformer-based language models [10][11] form the basis of this study. A transformer is a neural architecture noted mainly for its self-attention mechanism, which allows it to detect how each token in a query affects every other, and consequently which are most important in text generation. Self attention and multi-head attention supports detecting higher-order relational dependencies that are essential to analogy formation — for example, recognizing that the relation between pilot and airline mirrors the relation between sailor and ship. Unlike recurrent models [12], which process input sequentially and often struggle with long-range dependencies, Transformers compute global contextual relationships in parallel, enabling the efficient capture of abstract relational structures.

Transformers also produce contextual embeddings — vector representations whose meanings are conditioned on surrounding words. Because analogies require models to understand a word not in isolation but in relation to others, contextualized vectors are theoretically well suited for capturing the structure of analogical pairs. Multi-head attention expands this capability by enabling the model to track different similarity aspects simultaneously.

Additionally, the emergent behavior of transformer-based models during large-scale pretraining has been shown to give rise to geometric regularities in embedding space, such as linear relational analogies, cluster structures, and directional semantic offsets. These geometric properties make Transformers a strong testbed for studying analogy reasoning because they allow analogical relationships to manifest in interpretable, measurable forms. Conducting a comparative evaluation across different

Transformer families — masked-language models, autoregressive models, and encoder-decoder architectures — therefore provides both practical and theoretical insight into how architectural and training differences influence their ability to represent and generalize relational patterns.

In this study, five transformer-based language models are considered: BERT [13], RoBERTa [14], DistilBERT [15], GPT-2 [16], and T5 [17], which come from different transformer families. See below Table 1 for model comparisons.

Model	**Type**	**How it Works**
BERT	Two-Way Encoder	Reads text in both directions; predicts masked words using full context.
DistilBERT	Compressed BERT	Smaller, faster BERT variant retaining most accuracy with fewer parameters.
RoBERTa	More trained BERT	Improved BERT trained longer with dynamic masking for higher accuracy.
GPT-2	Next-Word Predictor	Generates text left-to-right; predicts the next word instead of filling blanks.
T5	Text-to-Text	Solves tasks by converting them into text transformations using special span tokens.

Table 1 Comparison of Transformer-Based Language Models

BERT, RoBERTa, and DistilBERT all belong to the encoder-only masked-language model (MLM) family. These models in general process text bidirectionally, attending to both left and right context and are trained using the masked language modeling objective, which involves a percentage of tokens being replaced with a mask token (e.g., `[MASK]`) and the model learns to predict the missing word. They then output a probability distribution over the vocabulary for the masked position. BERT is the original benchmark model for MLM tasks and enables strong relational understanding. RoBERTa is a more robustly trained version of BERT with dynamic masking. DistilBERT is a more lightweight version of BERT, enabling faster inference. Because MLMs predict missing words, they are ideal for fill-in-the-blank type of analogy completion.

GPT-2 is from the autoregressive model (Left-to-Right Predictive Family). It uses unidirectional (causal) self-attention, meaning that each token attends only to previous tokens. GPT-2 is trained to predict the **next token**, rather than fill in masked positions., which can lead to more varied or less constrained interpretations of relational prompts.

T5 belongs to the encoder–decoder transformer family. It uses both encoder self-attention for understanding the full input prompt and decoder self-attention for generating predictions autoregressively. In addition, cross-attention is employed where the decoder attends to encoder representations. T5 frames all tasks—including analogy completion—in a unified “text-to-text” format. For analogy prompts, T5 is prompted using natural-language instructions such as *“Fill in the blank:”*.

1.3 Focus of the study

By selecting models from all different transformer families, the study achieves a wide coverage of all transformer reasoning strategies and allows comparisons across architectures to analyze impact on analogy performance. Understanding the key differences of these five models analyzed offers an explanation of their differing performances within the study.

The paper is organized as follows; in Section 2, we introduce the Related Work in the area of analogical reasoning coupled with LLM's as well as some gaps in current research and motivations that informed the current study. The methodology that governs the experiment and some preliminary contextual details are summarized in Section 3. Section 4 gives the details of the experimental procedure. Furthermore, in Section 5, the visuals obtained as a result of the experiment are displayed. Section 6 contains a discussion and elaboration of the results obtained and conclusions about what they mean. Finally, we conclude the study's revelations and indicate avenues for future work in Section 7.

2 Related Work

Analogical reasoning has been a longstanding benchmark for assessing both human cognition and the capabilities of language models. Several recent studies have investigated how well pre-trained models can perform analogy tasks, and what factors influence their success.

E-KAR: A Benchmark for Rationalizing Natural Language Analogical Reasoning [18] introduced a knowledge-intensive analogy dataset derived from real exam sources, such as the Chinese Civil Service Exams. The benchmark includes two tasks: (1) analogical question answering, and (2) free-text rationalization explaining why candidate answers are correct or incorrect. Their findings highlight that state-of-the-art models often perform well superficially but struggle to provide meaningful explanations, revealing a reliance on shallow cues rather than deep relational reasoning. This underscores the need for both explainability and evaluation on knowledge-heavy analogies, which aligns closely with domain-specific tasks such as job profiling.

Can language models learn analogical reasoning? [19] explored models trained on analogical tasks resembling human psychological tests. They showed that with appropriate training objectives, models can approach human performance, achieving accuracy around 0.69. However, the study mainly uses small datasets and relatively simple analogy types, leaving open questions about performance on domain-specific analogies and about why models succeed or fail, which ties directly into interpretability concerns.

Evaluating the Robustness of Analogical Reasoning in Large Language Models [20] examined the sensitivity of GPT-family models to out-of-distribution variants, including paraphrasing, answer order changes, and other modifications that preserve the underlying reasoning. Their work demonstrates that even minor surface changes can significantly degrade model performance, whereas humans remain robust.

This highlights the importance of assessing not only accuracy but also robustness and generalization, particularly for models deployed in real-world tasks.

Similarly, **Using Counterfactual Tasks to Evaluate the Generality of Analogical Reasoning in Large Language Models** [21] created counterfactual analogy problems, explicitly designed to be unlike anything in the models' training data. Human performance remained high, while models' performance dropped substantially, indicating reliance on memorization or data overlap rather than abstract relational reasoning. These results motivate the use of paraphrase and counterfactual variants in domain-specific analogy evaluation and suggest integrating interpretability tools (attention, saliency, SHAP) to understand model reasoning.

2.1 Gaps and Motivation for Current Work

While prior studies provide valuable insights into model accuracy, knowledge dependency, and robustness, several limitations remain. Few works comprehensively evaluate multiple model types (masked, autoregressive, and encoder-decoder) on a small, domain-specific analogy dataset. There is also a lack of analysis comparing masked language models (BERT, RoBERTa, DistilBERT) with generative models (GPT-2, T5) in terms of both prediction accuracy and relational structure.

Our work addresses these gaps by conducting a comparative analysis of five pre-trained models on a curated analogy dataset focused on different analogy types. We combine top-k prediction analysis and other universally usable quantitative metrics to not only test whether models answer analogies correctly but also their ability to handle different analogy structures.

3 Methodology

This methodology explains the components involved in the method followed by the study.

3.1 Overview

This study investigates how five transformer-based language models—BERT, RoBERTa, DistilBERT, GPT-2, and T5—interpret and complete natural-language analogies. The study is designed to provide (1) a fair and standardized evaluation pipeline across differing model architectures, and (2) quantitative metrics that reflect reasoning quality. All experiments were conducted using Python (any version of Python3) and the HuggingFace *Transformers* library.

3.2 Dataset

3.2.1 Analogy Entries

The analogy dataset used for the study consists of **90 analogy questions** *structured specifically to each model* but generally in the form:

A is to B just like C is to D.

with one letter omitted at a time. As a result, the dataset contains analogies with missing tokens at the beginning, middle and end of the analogy to observe the effect of the different omissions in analogy completion performance. Analogies were manually constructed to cover a range of analogy categories, including:

- Functional analogies (e.g. "A pilot operates a plane in the air in the same way that a sailor operates a ship on water")
- Taxonomic analogies (e.g. "A sparrow is a type of bird in the same way that a salmon is a type of fish.")
- Role–object analogies (e.g. "A chef works in a kitchen in the same way that a teacher works in a classroom.")

The dataset contains prompts customized to each type of model for each unique analogy - for example, for one functional analogy there are five entries, one for each type of model. As a result, there are 9 unique types of analogy in the dataset: for each of the three types of analogy there are examples for each of the three omitted token positions.

Below is the breakdown of the dataset:

For each analogy type: 30 entries, 10 per omitted token position (beginning, middle, end). Each entry has 5 customized prompts for each model.

For example, for the functional analogy type with the omitted [MASK] token at the beginning, the following prompts are given to each model:

Model	**Prompt (Omitted Token at Beginning)**
BERT	[MASK] works for a airline, just like a sailor works for a ship.
DistilBERT	[MASK] works for a airline, just like a sailor works for a ship.
RoBERTa	<mask> works for a airline, just like a sailor works for a ship. .
T5	fill: <extra_id_0> works for a airline, just like a sailor works for a ship.
GPT-2	Sailor works for a ship. Similarly, a airline is worked at by a
Correct Answer	Pilot

Table 2 One functional analogy (mask at the beginning) with model-specific prompts and the correct answer.

Each unique analogy entry also has the corresponding correct answer (s) to check the model predictions against, the prediction of which form the foundation for the analysis. Although the dataset is relatively small, it is balanced across categories and serves as a controlled test set.

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3.3 Specific Models Under Study

As aforementioned, each of the five models under study will require changes made to the formatting of the dataset. For generative models (GPT-2, T5), generation is restricted to 1–2 new tokens, ensuring comparability with single-token MLM predictions. Five transformer models are included to represent the three major architectural families:

3.3.1 Masked Language Models (MLMs)

BERT-base-uncased, RoBERTa-base, and DistilBERT-base-uncased are the masked language models used in this study. These models operate by masking a missing token (e.g., `"[MASK]"` or `"<mask>"`) and predicting it using bidirectional self-attention. For these models, the dataset analogies will include a [MASK] token as a fill in the blank. For example: "Pilot is to Airline as Sailor is to [MASK]."

3.3.2 Autoregressive Model (AR)

GPT-2

This model predicts the next token in a left-to-right sequence, requiring a prompt structured as a continuation rather than a mask. For this model, the dataset will be written as a sentence, for example: "Pilot is to Airline as Sailor is to "

3.3.3 Encoder–Decoder Model

T5-Small

T5 reformulates all tasks as text-to-text transformations and fills blanks using special sentinel tokens (e.g., `<extra_id_0>`). The dataset will have to be formatted similarly to the MLM's, where an extra token will need to be included. For example: "Fill in the blank: Pilot is to Airline as Sailor is to ¡extra_id_0¿."

These families differ substantially in training objectives and attention flow, making them ideal for comparative analysis, although a noticeable difference in performance based on architecture differences and priorities is expected.

3.4 Preprocessing and Output Normalization

A few measures are employed to ensure fair scoring across models, such as token-level cleaning, which includes removing characters and stripping punctuation and converting to lowercase. Lemmatizing is also done where words are reduced to their lemma (e.g., "ships" → "ship"). For synonym mapping, WordNet similarity is used to treat synonyms (e.g., "vessel" for "ship") as correct, with configurable equivalence thresholds. Lastly, if a model outputs multi-word answers, the first content word is extracted for scoring. Full sequences are still used for semantic-similarity metrics.

3.5 Evaluation Metrics

To ensure that all five transformer models can be compared fairly, this study uses only metrics that every model can provide regardless of architecture (masked, autoregressive, or encoder-decoder). All metrics depend solely on the final predictions and the token-probability distribution, which are produced consistently across models. The model's prediction for the different positions of the omitted tokens are compared to the correct answers in order to form the analysis. The following metrics are elaborated upon below: Top-1 and Top-10 accuracy [22], confidence [23] and semantic similarity [24].

3.5.1 Top-1 Accuracy

This statistic measures whether the model's highest-confidence prediction exactly matches the expected answer or an accepted synonym. In the resuls the Top- 1 Accuracy is averaged over the different analogy types and omitted token positions to get a overall view of a specific model's performance.

$$\text{Top-1 Accuracy} = \begin{cases} 1, & \text{if the prediction is correct} \\ 0, & \text{otherwise} \end{cases}$$

3.5.2 Top-10 Accuracy

This statistic measures whether the model's highest-confidence prediction exactly matches the expected answer or an accepted synonym. In the resuls the Top-10 Accuracy is averaged over the different analogy types and omitted token positions to get a overall view of a specific model's performance.

$$\text{Top-10 Accuracy} = \begin{cases} 1, & \text{if the cprrect prediction is within top 10 predictions} \\ 0, & \text{otherwise} \end{cases}$$

3.5.3 Prediction Confidence

This measure is the probability assigned to the model's top prediction (range 0-1). This metric reflects how "sure" the model is, regardless of correctness that the prediction it made is correct.

3.5.4 Semantic Similarity Score

Uses WordNet path similarity between the predicted answer and the correct answer (range 0–1).

This captures partial correctness—e.g., predicting *"vessel"* instead of *"ship"*.

4 Experimental Procedure

This section details the exact experiment conducted and the code segments for that step.

4.1 Step 1 — Standardized Prompting

Each analogy is converted into a model-appropriate format but uses the same conceptual input.

Examples for omitted token at the end for any analogy type:

Model	**Input Format**
BERT / DistilBERT	*"... works for a [MASK]."*
RoBERTa	*"... works for a ¡mask¿."*
GPT-2	*"... works for a "* (model continues the sentence)
T5	*"Fill in the blank: ... works for a ¡extra_id_0¿."*

Table 3 Model-specific input formats used for analogy completion.

Each model produces a single "best guess" to the analogy, as well as the next 10 predictions and associated probabilities.

The dataset creation is in the `dataset.py` file and contains the automated generation of analogies.

```
for idx, (a, a_place, b, answers) in enumerate(functional_examples,
↪  1):

    add_analogy(
        id=f"functional_{idx:03d}_beginning",
        analogy_type="functional",
        mask_position="beginning",
        correct_answers=[a],
        bert_prompt=f"[MASK] works for a {a_place}, just like a {b}
        ↪  works for a {answers[0]}.",
        distil_prompt=f"[MASK] works for a {a_place}, just like a {b}
        ↪  works for a {answers[0]}.",
        roberta_prompt=f"<mask> works for a {a_place}, just like a {b}
        ↪  works for a {answers[0]}.",
        t5_prompt=f"fill: <extra_id_0> works for a {a_place}, just
        ↪  like a {b} works for a {answers[0]}.",
        gpt2_prompt=f"{b.capitalize()} works for a {answers[0]}.
        ↪  Similarly, a {a_place} is worked at by a"
    )
```

The above segment shows the automated creation of the functional analogy type with the omitted token in the *beginning* position. The `add_analogy()` function further generates all omitted-token positions for all analogy types (9 function calls in total).

The `functional_examples` variable is a dictionary such as follows:

```
    functional_examples = [
    ("pilot", "airline", "sailor", ["ship", "boat", "vessel"]),
```

```
    ("chef", "kitchen", "teacher", ["school", "classroom"]),
    ("driver", "car", "captain", ["ship", "boat"]),
    ("writer", "pen", "artist", ["brush", "paint"]),
    ("farmer", "field", "fisherman", ["river", "boat"]),
    ("doctor", "hospital", "nurse", ["hospital", "clinic"]),
    ("judge", "court", "lawyer", ["courtroom", "chamber"]),
    ("pilot", "plane", "astronaut", ["spaceship", "rocket"]),
    ("musician", "instrument", "singer", ["microphone", "stage"]),
    ("coach", "team", "teacher", ["class", "students"])
]
```

The components `a`, `a_place`, `b` and `answers` can be found in each row of this dictionary corresponding to the position. See below for an example from the first row of `functional_examples`

Variable	Value (from Row 1)
`a`	pilot
`a_place`	airline
`b`	sailor
`answers`	{ship, boat, vessel}

Table 4 Illustration of variables `a`, `a_place`, `b`, and `answers` used in constructing functional analogy prompts.

A similar logic is applied for all other analogy types and omitted token positions.

4.2 Step 2 — Output Normalization

To make all predictions comparable:

1. Remove tokenization artifacts (##, Ġ).
2. Convert to lowercase.
3. Keep only alphanumeric characters.
4. If multi-word outputs occur, keep the **first meaningful word**.
5. Lemmatize using WordNet.

This guarantees a single standardized answer from each model.

4.3 Step 3 — Compute Evaluation Metrics

For each model × prompt pair, the following metrics are computed:

4.3.1 Top-1 accuracy

The following function is used to calculate the top-1 accuracy of each model's predictions. `1` is returned if the first prediction matches the set of correct answers for that specific prompt, and `0` otherwise.

```
def top1_accuracy(preds, correct_answers):
return 1.0 if preds and preds[0].lower().strip() in correct_answers
↪  else 0.0
```

4.3.2 Confidence

This section highlights how confidence was calculated for the five different models.

BERT, DistilBERT, and RoBERTa

The below code segment explains how confidence is calculated for the three masked language models in the study.

```
mask_index = (input_ids ==
↪  mask_token_id).nonzero(as_tuple=True)[1].item()
logits = outputs.logits[0, mask_index]
probs = torch.softmax(logits, dim=-1)
top_prob, top_id = torch.max(probs, dim=-1)
confidence = top_prob.item()
```

Since these three are masked language models, they produce a probability distribution only over the masked position. The confidence is simply the softmax probability of the top token at that position.

GPT-2

GPT-2 is a left-to-right autoregressive language model, so it predicts the next token after the prompt, as can be seen from the below code. The confidence is the probability of the next-word prediction.

```
with torch.no_grad():
    logits = gpt2_model(**gpt_inputs).logits
    next_logits = logits[0, -1]
    next_probs = softmax(next_logits, dim=-1)
_, top_indices = torch.topk(next_probs, 10)
gpt2_preds = [gpt2_tokenizer.decode([idx]).strip(string.punctuati ⌋
↪  on) for idx in top_indices]
gpt2_confs = [float(next_probs[idx]) for idx in top_indices]
```

T5

T5 is an encoder–decoder generator, not a masked labguage model, so there is no mask token; instead, T5 generates a span where <extra_id_0> was. The confidence is taken from the **first decoder timestep,** which corresponds to the “first meaningful word of the answer.”

```
    outputs = model.generate(
    input_ids,
    num_beams=10,
    output_scores=True,
    return_dict_in_generate=True
)

# Identify the first generated token after <extra_id_0>
first_token_id = outputs.sequences[0][1].item()

# Score of that step
logits = outputs.scores[0][0]       # first decoder timestep, beam 0
probs = torch.softmax(logits, dim=-1)
confidence = probs[first_token_id].item()
```

4.3.3 Semantic Similarity

The following function computes semantic similarity as cosine similarity in the code. It cycles between all accepted correct answers and returns the highest similarity of the predicted word to the answer.

```
def semantic_similarity(model, tokenizer, word, correct_answers):
    """Cosine similarity between a predicted word and the closest
    ↪  correct answer."""
    sims = []
    for ans in correct_answers:
        print(ans)
        inputs = tokenizer([word, ans], return_tensors="pt",
        ↪  padding=True, truncation=True)
        with torch.no_grad():
            emb = model(**inputs).last_hidden_state.mean(dim=1)
        sim = F.cosine_similarity(emb[0].unsqueeze(0),
        ↪  emb[1].unsqueeze(0))
        sims.append(sim.item())
    return max(sims)
```

This yields a dataset for statistical comparison.

4.4 Step 4 — Aggregate Results

Metrics are aggregated:

- By model (overall performance)
- By analogy type (functional / taxonomic / role / relational)
- Across paraphrase prompts (robustness)

This allows both overall rankings and category-specific insights.

5 Results

5.1 Average Over Analogy Types: Metrics Across Models Given MASK Position

This section explores the average of the metrics across the different analogy types considered in the study (functional, role-object, taxonomic). The following line plots provide insight into the behavior of the models over certain metrics overall, and allow the focus to be on comparing performance given MASK position in general.

The following line plot shows the Top-1 accuracy of different models by MASK position overall. As can be seen, most models improved their Top-1 accuracy when the omitted token was at the end, rather than the beginning, so an upward trend can be seen for most models, except for T5, which peaks at the MASK position being in the middle. It can also be observed that the RoBERTa model had the greatest Top-1 accuracy overall, with BERT also matching its performance when the MASK token is at the end.

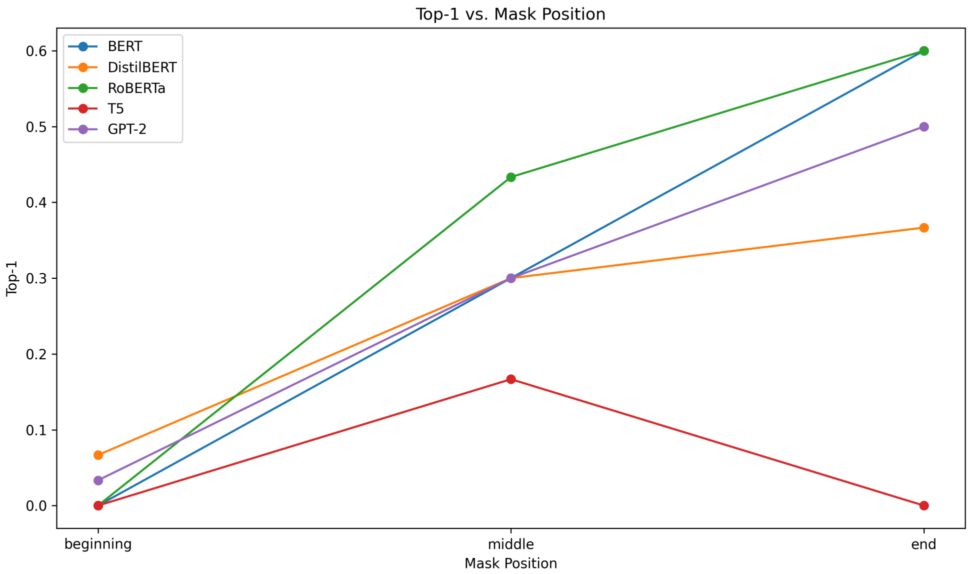

Fig. 1 Top-1 Across Analogy Tasks and Models

Figure 2 shows the semantic similarity of the models' predictions to the accepted right answers overall, by MASK position. it can be seen that a similar trend to Figure 1 can be observed in which most models performed best (achieved greatest semantic similarity) when the MASK token was at the end, with the exception of T5 which once again peaked at MASK position middle. Although an interesting phenomenon can be observed in which T5 is among the best models semantic similarity wise when it comes to MASK token beginning, yet the worst when it comes to MASK token end. A familiar trend can also be observed in that the three BERT-based models are highest performing overall.

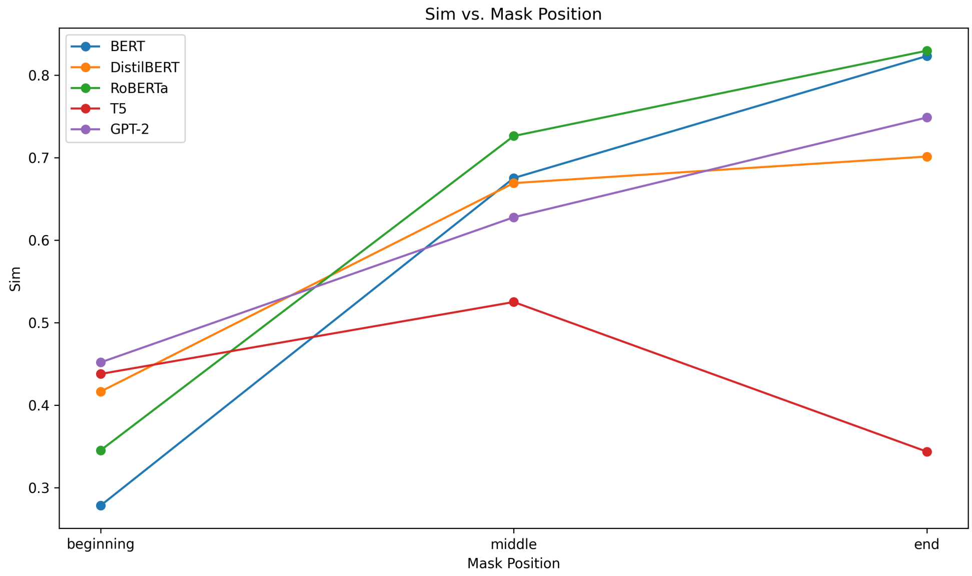

Fig. 2 Semantic Similarity Across Analogy Tasks and Models

Figure 3 shows the confidence scores of each model's predictions across the three MASK positions. Because raw confidence values are not directly comparable across different architectures—each model produces probabilities under different vocabularies, objectives, and calibration behaviours—the figure should be interpreted primarily in terms of relative changes across MASK positions within each model, rather than absolute differences between models. Even under this restricted interpretation, a pattern similar to earlier results can be observed: most models tend to reach their highest confidence when the MASK token appears in the middle or end of the input rather than the beginning. Notably, both RoBERTa and T5 peak at MASK position middle, with T5 showing its largest internal confidence jump at that point. BERT, in contrast, exhibits a relatively stable profile with only slight variation across positions. DistilBERT displays a moderate rise from beginning to middle before dropping again at end, making it the only model with a non-monotonic pattern across all three positions. Overall, the figure highlights how each model's internal certainty shifts depending on MASK placement, with encoder-style models generally showing more stable or favourable confidence patterns compared to GPT-2, while also illustrating distinctive behaviours such as T5's strong preference for MASK tokens appearing in the middle of the input.

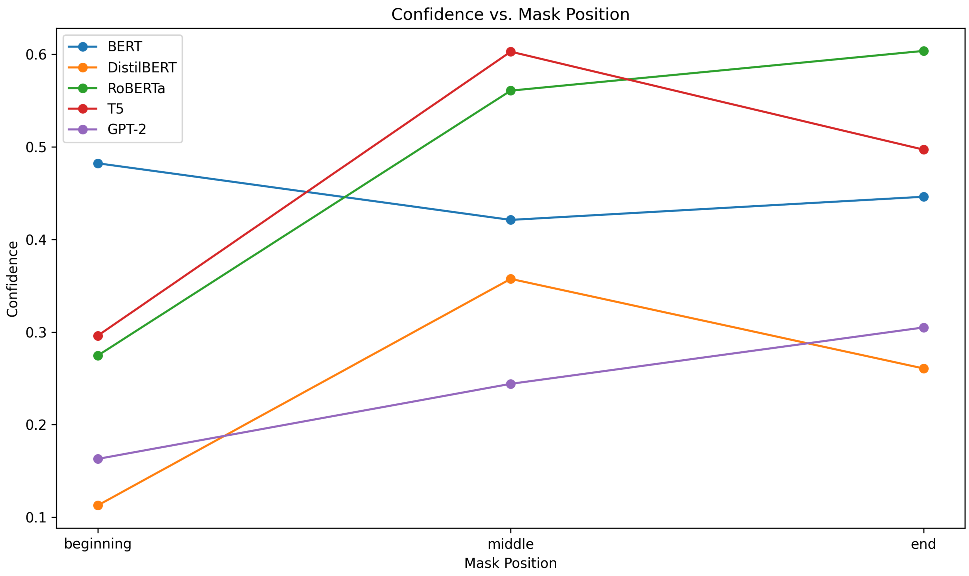

Fig. 3 Confidence Across Analogy Tasks and Models

5.2 Detailed Breakdown of Analogy Types: Metrics Across Models and MASK Position

This section contains heatmaps that represent the metrics per model per analogy type by MASK position. There are three figures, one for each MASK position. The plots below provide information about the comparisons between the performance of the models when it comes to MASK positions.

For Top-1 accuracy over MASK positions, the models performed best when the MASK token is in the end, closely followed by MASK position at the middle. Differing analogy types did not lead to any major differences in performance, and the most interpretable analogy type to the models changed every MASK position. For example, from Figure 4 it can be seen that the taxonomic analogy type was most difficult for the models when MASK position was at the beginning, but was best performing when MASK position was in the middle. The three BERT-based models were the highest performing in terms of Top-1 accuracy.

Top-10 accuracy was broader in including all analogy types, but an interesting effect can be observed for the T5 model at the MASK position at the end - it had very low Top-1 and Top-10 accuracy when the mask position was at the end, however it performed better when MASK position was at the beginning and middle, contrary to the other models in the study.

Semantic similarity also increases across MASK position for all models except T5, whose predictions become less similar to the expected answers at the end. Confidence scores will not be compared against each model but can be compared across MASK positions - and a great difference between middle and end MASK positions is not displayed. Though it should be noted that RoBERTa had higher confidence scores for the middle and end MASK positions than its masked language model members (BERT and DistilBERT).

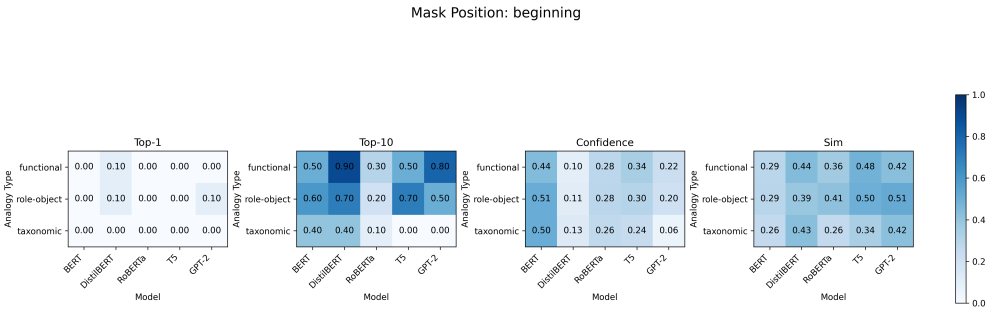

Fig. 4 Metrics Across Models for [MASK] Beginning Position

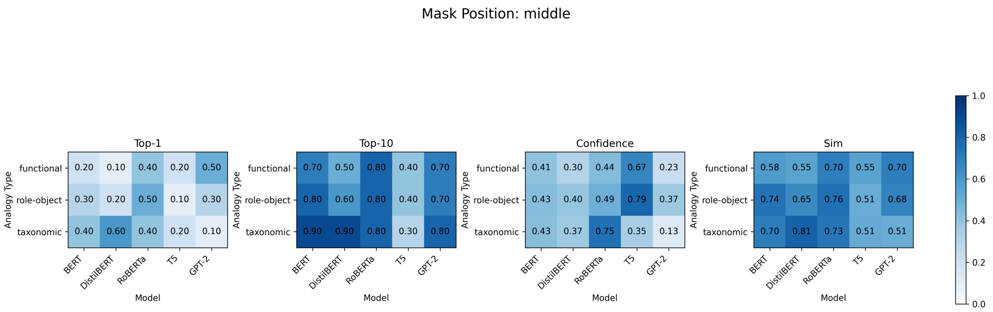

Fig. 5 Metrics Across Models for [MASK] Middle Position

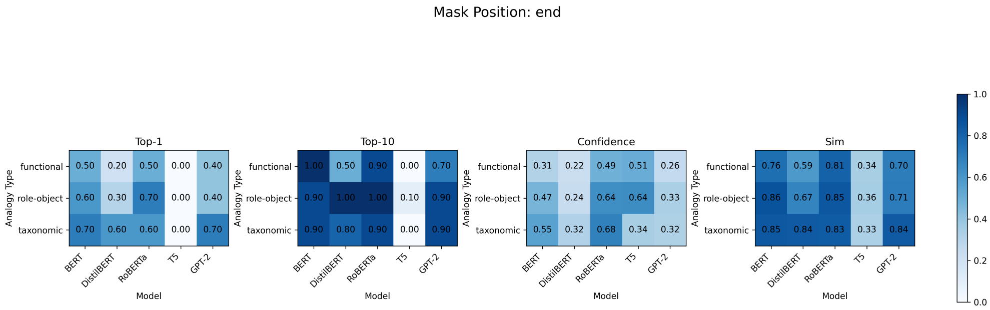

Fig. 6 Metrics Across Models for [MASK] End Position

5.3 Overall Model Profiles

This section contains the radar plots of all models, so as to best get a general snapshot of analogy performance overall. It is important to note that these are average metrics across MASK positions and analogy types, so some metrics may appear lower than in previous figures due to them being averaged. These are the most general results of the study.

As can be seen from the following figures, all models except for T5 had a very high Top-10 metric, indicating that the correct analogical answer was highly likely to be in the Top-10, for most models except T5. Semantic similarity across models is also high over all models except for T5 also. The Top-1 metric was between 0.2 and 0.35 for all models except T5 which was around 0.05 overall.

For all models, Top-10 was higher than Top-1, indicating that while the correct prediction was not the first predicted answer, it was found in subsequent predictions more often. Overall, RoBERTa performed the best in all metrics considered, while it can be seen that T5 had the worst overall performance.

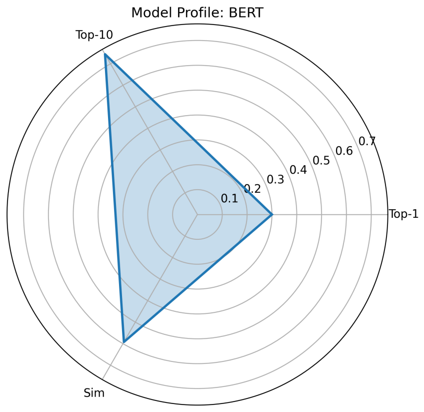

Fig. 7 BERT Model Profile

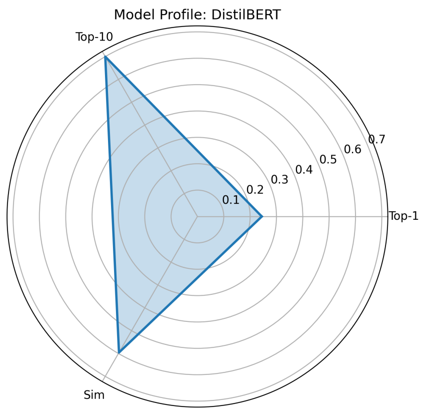

Fig. 8 DistilBERT Model Profile

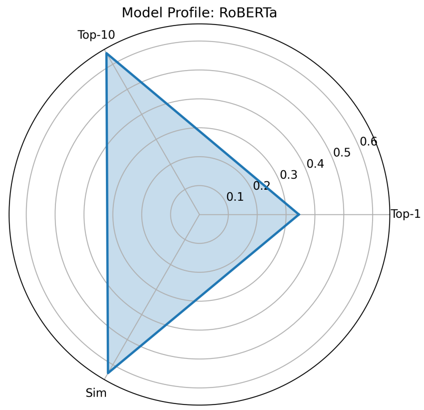

Fig. 9 RoBERTa Model Profile

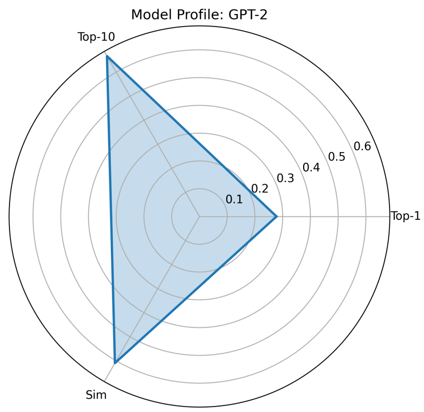

Fig. 10 GPT-2 Model Profile

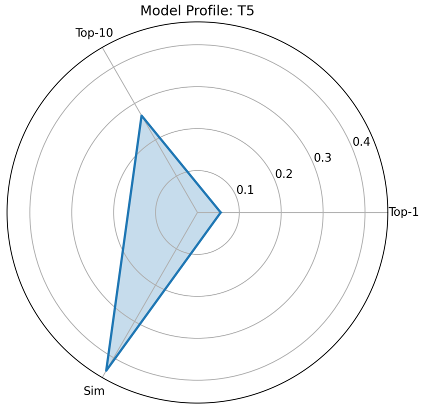

Fig. 11 T5 Model Profile

6 Discussion

6.1 Overview

The results of this study reveal clear and consistent performance differences across transformer architectures when applied to analogical reasoning tasks. Among the evaluated models, RoBERTa emerged as the strongest performer across all evaluation metrics, while T5 was the weakest, particularly in accuracy-based and semantic similarity-based measures. These outcomes show the strong influence of model architecture and pre-training objectives on analogical reasoning capabilities, since these differ between RoBERTa and T5.

6.2 RoBERTa: Superior Performance in Analogy Reasoning

RoBERTa demonstrated the highest overall performance across Top-1 accuracy, Top-10 accuracy and semantic similarity, although DistilBERT and BERT were close behind. This superiority can be attributed to several architectural and training-related factors.

First, RoBERTa's masked language modeling objective, combined with its dynamic masking strategy, appears particularly well suited to analogy completion tasks. Unlike static masking approaches, dynamic masking exposes RoBERTa to a wider variety of masked contexts during training, enabling more flexible contextual inference. This likely enhances the model's ability to infer relational structure when predicting a missing term within an analogy.

Second, RoBERTa benefits from larger-scale pretraining on more diverse corpora and longer training schedules compared to BERT. These enhancements have been shown to improve contextual sensitivity and semantic precision, which are essential to identify the correct analogical correspondences. The consistently higher confidence scores observed for RoBERTa suggest that the model not only predicts correct answers more frequently, but also assigns higher probability mass to those predictions, indicating stronger internal certainty.

Finally, RoBERTa's strong semantic similarity scores indicate that its predictions remain closely aligned with the intended relational meaning of the analogy. This suggests that RoBERTa encodes relational information at a deeper semantic level, rather than relying solely on surface-level co-occurrence statistics.

Nevertheless, the strong performance of all three masked language models reinforces the conclusion that encoder-only architectures are well suited for structured analogy completion, especially when the task can be framed as a missing-token prediction problem.

6.3 Limitations of T5 in the Analogy Setting

In contrast, T5 exhibited the weakest performance across nearly all metrics, including Top-1 accuracy, Top-10 accuracy, and semantic similarity. This underperformance is particularly notable given T5's strong performance on many text-to-text benchmarks.

One primary explanation could lie in T5's encoder–decoder architecture and text-to-text training paradigm, which differs fundamentally from masked language modeling. T5 is trained to generate entire output sequences rather than predict a single missing token. As a result, when constrained to produce short span-based outputs for analogical reasoning, T5 may fail to align its generation process with the evaluation framework used in this study.

Additionally, T5's outputs often consisted of longer phrases or reformulated sentences rather than concise, single-token answers. While such behavior may reflect deeper generative reasoning, it is penalized under token-level evaluation metrics. This mismatch between T5's generation style and the task formulation likely contributes to its low accuracy and semantic similarity scores.

6.4 Implications for Model Selection in Analogy Tasks

The contrast between RoBERTa and T5 underscores an important methodological insight: the effectiveness of a language model in analogy reasoning depends not only on model size or architecture, but also on alignment between training objectives and task formulation.

RoBERTa's success suggests that masked-token prediction remains a highly effective proxy for evaluating analogical reasoning, particularly in structured and constrained settings. Conversely, T5's performance indicates that generative, instruction-based models may require alternative evaluation strategies, richer prompting, or fine-tuning to demonstrate their full reasoning potential.

7 Conclusion

This study presented a systematic comparative analysis of transformer-based language models on analogical reasoning tasks, focusing on five representative architectures:

BERT, DistilBERT, RoBERTa, GPT-2, and T5. By evaluating all models under a unified experimental framework and shared evaluation metrics, the study aimed to isolate how architectural design and training objectives influence analogical reasoning performance.

The results demonstrate that masked language models outperform autoregressive and encoder–decoder models on analogy completion tasks formulated as missing-term prediction. Among all evaluated models, RoBERTa consistently achieved the strongest performance, attaining accuracy across the evaluated analogies while also exhibiting higher confidence and stronger semantic alignment with the answers. These findings suggest that RoBERTa's enhanced pretraining strategy—particularly dynamic masking and large-scale corpus exposure—enables more robust encoding of relational semantics.

In contrast, T5 exhibited the weakest performance across all metrics, underscoring a mismatch between sequence-to-sequence generation objectives and the constrained evaluation setup used in this study. While T5 excels in generative and instruction-based tasks, its difficulty in producing concise, single-token answers limited its effectiveness under token-level evaluation. GPT-2 similarly struggled to consistently produce correct analogical completions, reflecting the challenges faced by autoregressive models when precise relational inference is required rather than open-ended continuation.

Overall, this study reinforces the importance of task–architecture alignment when evaluating reasoning capabilities in large language models. The findings suggest that masked language models remain particularly well suited for structured analogy reasoning tasks, while generative models may require alternative prompting strategies or evaluation methodologies to fully demonstrate their reasoning potential.

Future research can extend this work by expanding the size and diversity of the analogy dataset, exploring multi-word and cross-domain analogies, and incorporating human judgment into semantic evaluation. Additionally, evaluating newer instruction-tuned and hybrid models may provide further insight into whether advances in prompt-following and reasoning alignment can bridge the observed performance gap. By continuing to refine both experimental design and evaluation metrics, future studies can deepen our understanding of how different transformer architectures encode and apply relational knowledge.

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